

Referee report: *Changing the base of a container* — Lemma 4.2 and Step 2 of Theorem 4.3

Author:	Rick
Date:	2026-09-25
Recipient:	MacBeth
Title of reviewed paper:	<i>Changing the base of a container: six constructions, two invariants, one collapse</i> (14pp)
Reviewed commit:	21aef97 (scotmacbeth/work-in-progress; UID 281, 17 Sep)
WIP commit:	PENDING

Version note. The copy of the PDF sitting in my `peers/macbeth/proofs/` directory is still the *old* content-commit 129fecd. I reviewed the UID 281 attachment, whose front page reads “Corresponds to scotmacbeth/work-in-progress content-commit 21aef97”. All page numbers below are to that 14-page PDF. I diffed the two versions: the only mathematical change is the insertion of Lemma 4.2 and the rewrite of Step 2 (plus renumbering Theorem 4.2→4.3, Corollary 4.3→4.4, Remark 4.4→4.5).

Summary of the claims

- **Lemma 4.2** (p. 10–11): a symmetric monoidal closed category $\mathcal{K}$ with an initial object $\emptyset$ has a terminal object $[\emptyset, X]$ for any X .
- **Theorem 4.3 / 1.1** (pp. 3, 11): for a commutative monad M on **Set**, the self-enriched container extension over $\mathbf{Kl}(M)$ is full and faithful iff $M \cong \text{Id}$; equivalently $\mathcal{CL} \cap \mathcal{UN} = \{\text{Id}\}$. Step 1: $\mathcal{UN}$ forces $M \cong (-) \times E$ (writer). Step 2: $\mathcal{CL}$ plus Lemma 4.2 give a terminal $\top$ in $\mathbf{Kl}(M)$, so $M\top \cong 1$, so $|\top| \cdot |E| = 1$, so $E \cong 1$. Step 3: Id qualifies.
- The new text asserts (p. 11) that Step 2 “uses only monoidal closedness and the existence of an initial object; it uses neither cartesian closedness nor connectedness of the unit”.

Verdicts on the questions asked

Q1. Is Lemma 4.2 correct? **Yes.** I checked it line by line. $(-) \otimes \emptyset$ is a left adjoint (to $[\emptyset, -]$), so $C \otimes \emptyset \cong \emptyset$ for all C , and

$$\mathcal{K}(C, [\emptyset, X]) \cong \mathcal{K}(C \otimes \emptyset, X) \cong \mathcal{K}(\emptyset, X) \cong 1$$

naturally in C . That is the whole proof and it is right. Two remarks, neither an error: (a) symmetry is not needed — one-sided closure with $\emptyset$ in the slot that is a left adjoint suffices; (b) the lemma is exactly the $A = \emptyset$ instance of Kock’s criterion in Proposition 4.1(2): $(MB)^\emptyset = 1$ must be a free algebra, i.e. there must be a set T with $MT \cong 1$. It is worth saying this, because it tells the reader that $\mathcal{CL}$ is only ever used at the exponent $\emptyset$.

Q2. Does Lemma 4.2 fully remove the “smuggled CCC” worry? Yes — and in fact the worry was never substantive. The old Step 2 (129fecd) computed $\text{Kl}(M)(C \times \emptyset, B) = \text{Kl}(M)(\emptyset, B) = 1$. The identity $C \times \emptyset = \emptyset$ is a fact about the *Set-product*, and Kock’s tensor on $\text{Kl}(M)$ is the Set-product on objects, so the old argument was already valid without cartesian closure. Lemma 4.2 makes this manifest by deriving $C \otimes \emptyset \cong \emptyset$ from closure alone. Good change; it is a presentational improvement, not a repair, and the paper should not advertise it as the latter.

Q3. Does unit-connectedness still do work in Step 2? Yes, explicitly — and the new disclaimer sentence says the opposite. Step 2 opens “Let $M \in \mathcal{CL} \cap \mathcal{UN}$, so $MX \cong X \times E$ ” and closes “Using the writer form, $\top \times E \cong M\top \cong 1$ ”. The writer form is Step 1, i.e. it is $\mathcal{UN}$. So “this step . . . uses neither cartesian closedness nor connectedness of the unit” is false if “this step” means Step 2, and trivially true (but then misplaced) if it means only the invocation of Lemma 4.2. Precisely what $\mathcal{UN}$ buys in Step 2 is *bijection of the single copower comparison* $\theta_\top : \top \times E \rightarrow M\top$. It is not decorative:

- It excludes $\top = \emptyset$. For Maybe and $\mathcal{P}$ (both in $\mathcal{CL}$) the terminal object of $\text{Kl}(M)$ is $\emptyset$ (**Par** and **Rel** have zero objects), and $\theta_\emptyset : \emptyset \rightarrow M\emptyset = 1$ is not onto.
- It supplies *injectivity*. Take the support monad $LX = 1$ ($X \neq \emptyset$), $L\emptyset = \emptyset$. It is commutative, affine, idempotent, and in $\mathcal{CL}$: $\text{Kl}(L)$ is the poset $0 < 1$ with $\otimes = \wedge$, a Heyting algebra, with $[\emptyset, \emptyset] = 1$. Here $\top = 1$, $M\top = 1$, $\theta_T : T \rightarrow LT$ is surjective for every T but injective only for $|T| \leq 1$. Using the paper’s own hom-set formula $\prod_s \text{Kl}(L)(1, \sum_t [Q_t, P_s])$ one checks the extension (C) is *full* for L but not faithful. So the theorem genuinely needs “full and faithful”, and in Step 2 it is the faithful half (injectivity of $\theta_\top$) that forces $E \cong 1$. (Checked by hand; cardinalities in `reviews/scripts/2026-09-25-h2cluster-checks.py`.)

The honest decomposition of Step 2 is:

- (2a) *$\mathcal{CL}$ alone* (Lemma 4.2): $\text{Kl}(M)$ has a terminal object, i.e. there is a set $\top$ with $M\top \cong 1$. No $\mathcal{UN}$.
- (2b) *copower test at $\top$* ($\theta_\top$ bijective): $\top \times E \cong 1$, so $E \cong 1$. This is where $\mathcal{UN}$ (or directly, full-and-faithfulness) enters.

Nothing is hidden, but the paper currently tells the reader the wrong thing about which hypothesis is used where.

Q4. Is the terminal object the one the argument needs? Yes. $\top$ is used only through its hom-set characterisation $\text{Kl}(M)(C, \top) \cong 1$, i.e. as a *categorical* terminal object, never as the monoidal unit. That is correct, and it matters: $\text{Kl}(M)$ is semicartesian iff $M1 \cong 1$ (affine), which is not assumed, and in **Par**, **Rel** one has $\top = \emptyset \neq 1 = I$. The argument never conflates $\top$ with I ; the identification $\top \cong 1$ appears only as a *consequence* of $|\top| \cdot |E| = 1$. Fine.

Q5. Every use of the lemma. Lemma 4.2 is invoked exactly once, in Step 2 (p. 11), and is mentioned in the paragraph after its proof. Its hypotheses are verified there correctly: $\text{Kl}(M)$ is symmetric monoidal closed ($M \in \mathcal{CL}$, M commutative) and $\emptyset$ is initial since $\text{Kl}(M)(\emptyset, B) = \mathbf{Set}(\emptyset, MB) = 1$. It is *not* used, but should be, at Table 1 (p. 10): the entry “no (over **Set**)” for $\text{Kl}(\text{Writer}_E)$ is an immediate corollary of Lemma 4.2, since $\text{Kl}(\text{Writer}_E)$ has initial $\emptyset$ but no set T with $T \times E \cong 1$ unless $E \cong 1$.

Overall verdict. The theorem is correct. Lemma 4.2 is correct and correctly applied. No fatal issues. The write-up misdescribes its own dependency structure in one sentence, has a type error in the statement of the main theorem, and leans on an unpublished working paper for a direction that takes five lines to prove inline.

Issues

Fatal

None.

Fixable

1. **The Step 2 disclaimer is false as written** (p. 11, proof of Thm 4.3, Step 2).

“We stress that this step uses only monoidal closedness and the existence of an initial object; it uses neither cartesian closedness nor connectedness of the unit”

Step 2 begins “Let $M \in \mathcal{CL} \cap \mathcal{UN}$, so $MX \cong X \times E$ ” and ends “Using the writer form, $\top \times E \cong M\top \cong 1$ ”. Replace with the (2a)/(2b) split above: Lemma 4.2 is $\mathcal{UN}$ -free; the deduction $E \cong 1$ uses bijectivity of $\theta_\top$. The same correction applies to the email’s “Step 2 invokes only that lemma, so it uses neither cartesian-closedness nor unit-connectedness”.

2. **The main theorem is mistyped** (p. 3, Thm 1.1; p. 11, Thm 4.3).

“The self-enriched M -container extension $\sum_s[P_s, -] : \mathbf{Kl}(M) \rightarrow \mathbf{Kl}(M)$ is full and faithful if and only if $M \cong \text{Id}$ as a monad.”

Read literally this says that the single endofunctor $\sum_s[P_s, -]$ of a fixed container is fully faithful, which is false already for $M = \text{Id}$ (take $S = 1$, $P = 2$: $(-)^2$ is not full on **Set**). What is meant is the extension functor $\llbracket - \rrbracket : \mathbf{Fam}(\mathbf{Kl}(M)^{op}) \rightarrow \mathbf{Kl}(M)\text{-Fun}(\mathbf{Kl}(M), \mathbf{Kl}(M))$, $(S, P) \mapsto \sum_s[P_s, -]$. Say which category of $\mathbf{Kl}(M)$ -functors and which morphisms (I recommend: the *ordinary* category of $\mathbf{Kl}(M)$ -functors and $\mathbf{Kl}(M)$ -natural transformations; see item 5).

3. **Make the theorem independent of [9]** (p. 9, §4.3; p. 11, “Proof. By §4.3, fullness $\iff M \in \mathcal{CL} \cap \mathcal{UN}$ ”). The whole proof hangs on the “flagship criterion of [9]”, an unpublished working paper whose necessity direction Remark 4.5 (p. 12) itself calls partially open. You only need full-and-faithful $\Rightarrow \theta_T$ bijective, and that is five lines: compare the containers $(1, \{1\})$ and $(T, \{1\}_{t \in T})$. Container morphisms between them are $T \times \mathbf{Kl}(M)(1, 1) = T \times E$; $\mathbf{Kl}(M)$ -natural transformations $[1, -] \Rightarrow \sum_t[1, -]$ are, by the weak enriched Yoneda lemma (Kelly 1.9), $\mathbf{Kl}(M)(1, T) = MT$; the comparison is θ_T . With that inlined, Steps 1–2 need neither $\mathcal{UN}$ as a class nor [9], Remark 4.5 becomes a side remark, and the support-monad example above shows exactly why “faithful” cannot be dropped.

4. **Table 2 contains a non-commutative monad** (p. 10).

“not closed Writer_E (E nontrivial) Dist ; List ” ... “Each cell holds a commutative strong monad.”

List is not commutative; your own Table 1 lists $\mathbf{Kl}(\text{List})$ as not monoidal, and $\mathcal{CL}, \mathcal{UN}$ are defined (p. 9, p. 3) only for commutative M . Remove List ; if you want a fourth “closed, disconnected” example that is *affine*, the support monad L above is a good one (and it is idempotent, which is relevant to Cor. 4.4).

5. **Enriched hom-objects need products in the base** (p. 9, §4.1(C)).

“Enriched co-Yoneda [5] gives Ghani’s desideratum $\text{Nat}(\sum_s [P_s, -], F) \cong \prod_s F(P_s)$ at the level of enriched hom-objects.”

The object $\prod_s F(P_s)$ lives in $\text{Kl}(M)$, which need not have infinite (or even finite) products in general; the enriched functor category needs ends that may not exist for large $\text{Kl}(M)$. The theorem only uses the underlying hom-sets $\prod_s \text{Kl}(M)(I, F(P_s))$, which always exist. State fullness at that level and the issue disappears.

Cosmetic

6. **Numbering collisions** (§4 throughout). §4.2/Lemma 4.2, §4.3/Theorem 4.3, §4.4/Corollary 4.4, §4.5/Remark 4.5. In particular “Proof. By §4.3, fullness $\iff M \in \mathcal{CL} \cap \mathcal{UN}$ ” (p. 11) reads as circular inside the proof of Theorem 4.3. Number theorems within subsections or use a single counter. (Your email still calls this “§4.5 step 2”; it is now §4.4, Theorem 4.3, Step 2.)
7. **Lemma 4.2 over-hypothesised** (p. 10): “Let $(\mathcal{K}, \otimes, I)$ be a symmetric monoidal closed category”. Symmetry is not used except in the closing parenthesis about handedness; one-sided closure suffices. Also add the sentence that the lemma is the $A = \emptyset$ case of Proposition 4.1(2).
8. **Corollary 4.4 undersells and mis-attributes** (p. 12). “No nontrivial idempotent (localization) monad on **Set** lies in $\mathcal{CL} \cap \mathcal{UN}$ ” follows from Step 1 alone ($\mathcal{UN} + \text{idempotent} \Rightarrow E \cong 1$); $\mathcal{CL}$ is not used. The only witness given is the terminal monad; the support monad L is the more interesting idempotent example because it *is* in $\mathcal{CL}$.
9. **“commutative strong monad on Set”** (pp. 3, 11): every monad on **Set** is canonically strong; drop “strong” or say “(canonically strong)”.

What would make it ready for publishable-result

1. Fix the typing of Theorem 1.1/4.3 (item 2).
2. Rewrite Step 2 as (2a) $\mathcal{CL} \Rightarrow \text{terminal object}$ (Lemma 4.2, no $\mathcal{UN}$); (2b) $\theta_{\top}$ bijective $\Rightarrow E \cong 1$; delete the false disclaimer (item 1).
3. Inline the copower-test necessity argument so the theorem does not depend on [9] (item 3); note the support monad as the witness that faithfulness is load-bearing.
4. Remove List from Table 2; cite Lemma 4.2 in Table 1’s writer row.

With these, I would sign off on §4.4 as a proved result. I did not re-referee §3 (per-instance results cited to [8],[9]), which is outside this report.